Interpolation Tests

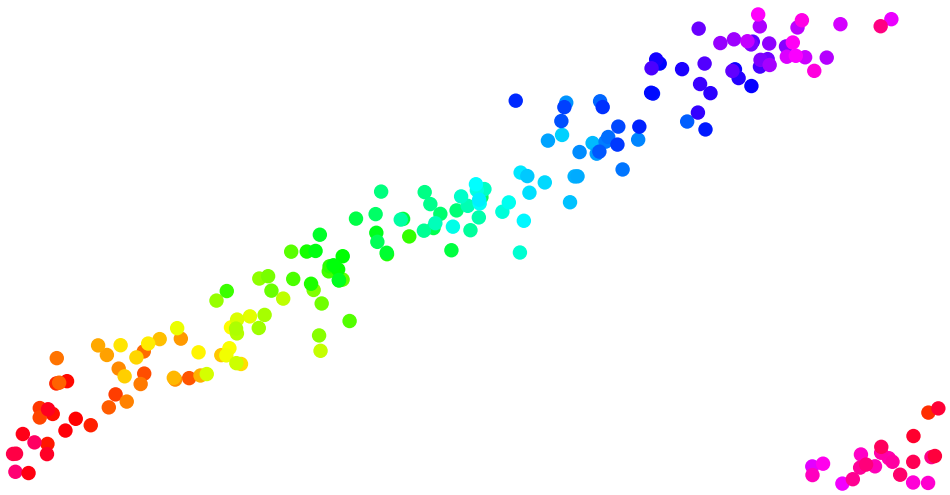
Michael Nisenzon

2023-03-14

7/4/2023

```
# Straight trends on (S^1)^2
set.seed(5)
n <- 200
samp_1 <- r_path_s1r(n = n, r = 2, k = c(1, 1), angles = TRUE)

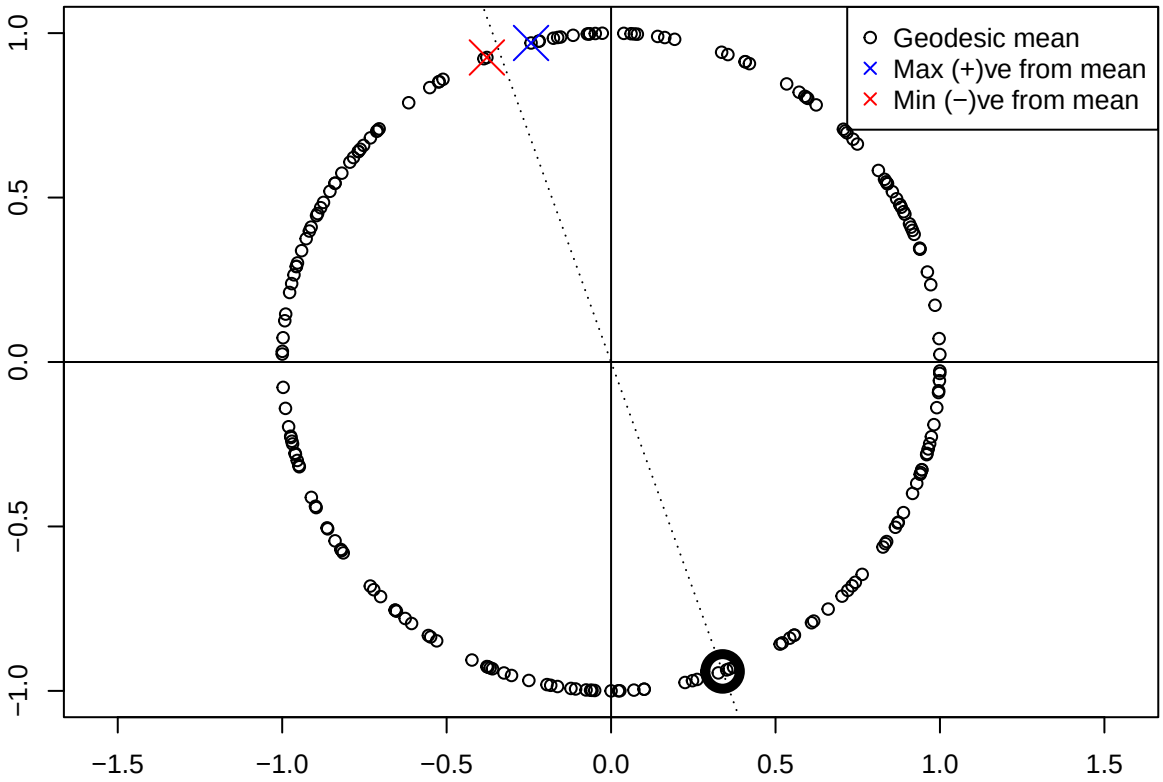
plot(samp_1,
  xlim = c(-pi, pi), ylim = c(-pi, pi), col = rainbow(n),
  axes = FALSE, xlab = "", ylab = "", pch = 16
)
```



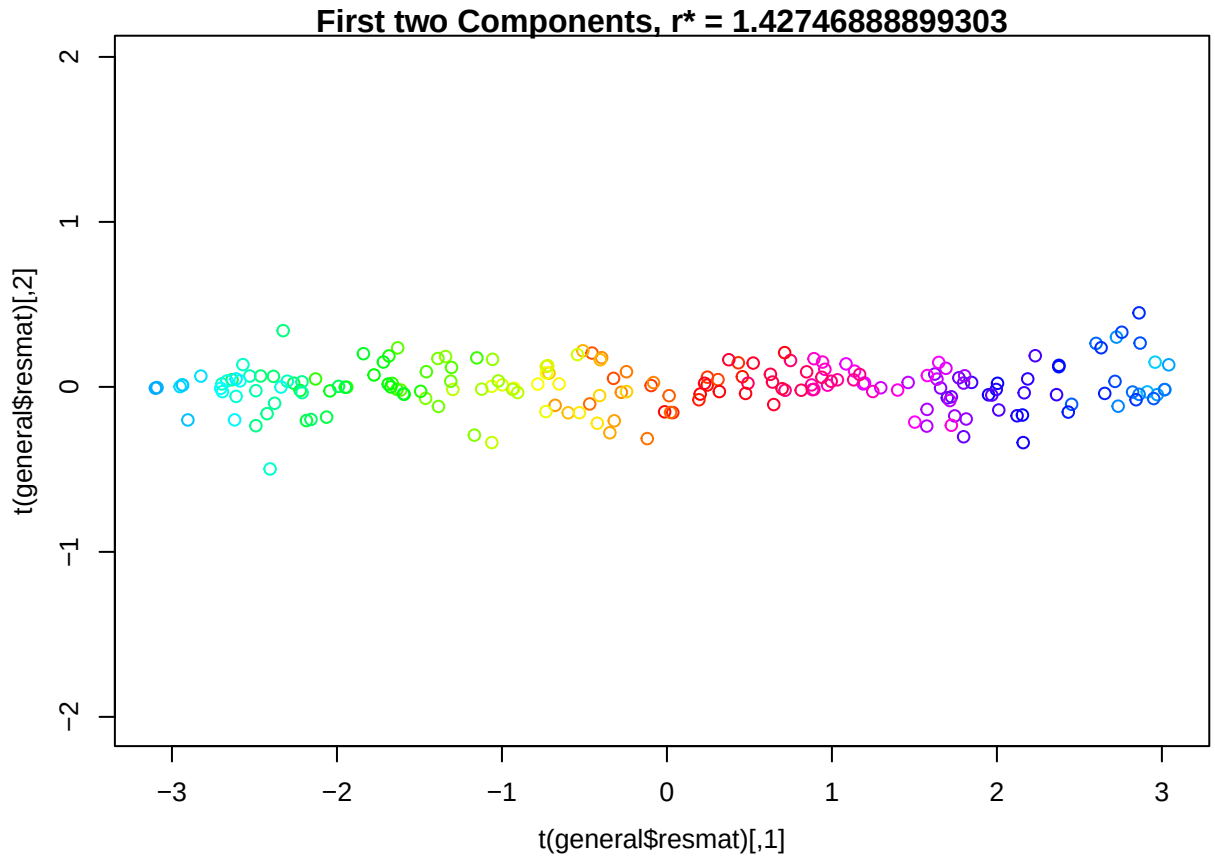
```
clusters <- samp_1

time <- Sys.time()
res_int <- plot_MDS_py(clusters, landmark_size = .1,tol=1e-3,landmark_tol=1e-3)
```

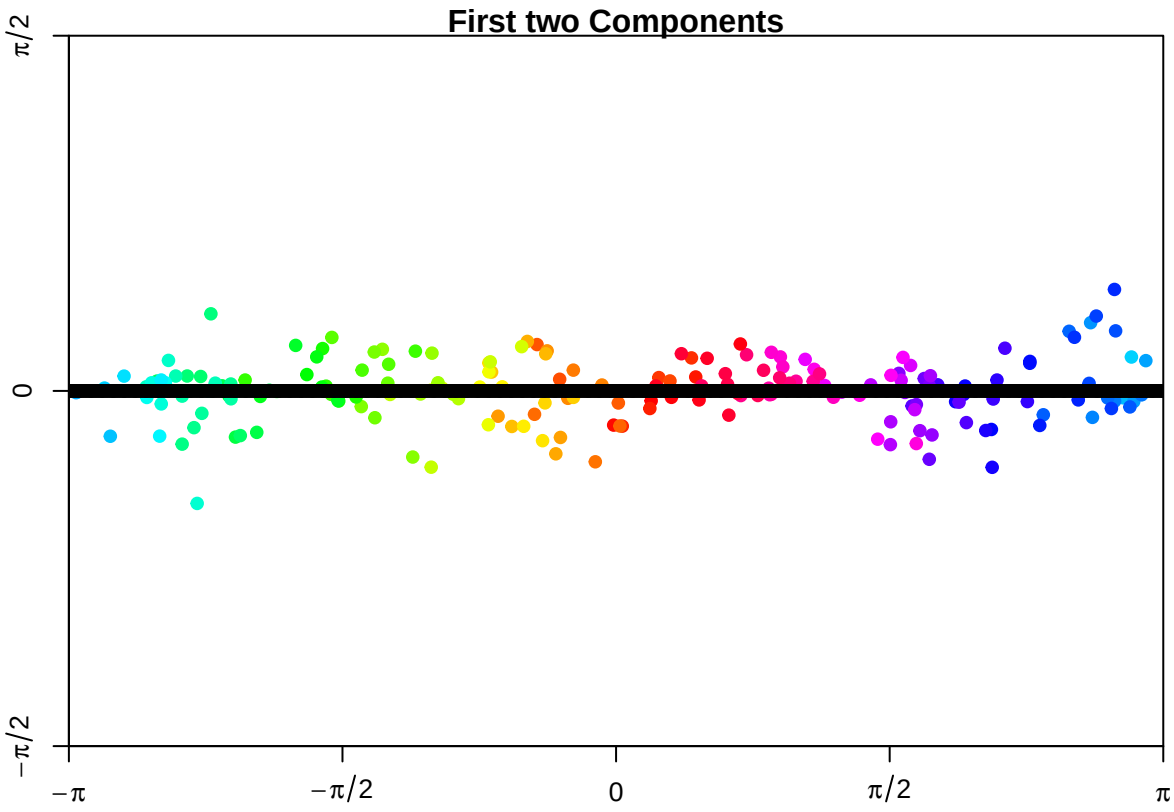
```
## Message from pns() : dataset is on 2-sphere.
##  .. sequential tests with significance level 0.1
## 2-sphere to 1-sphere, by GREAT sphere, p(LRT) = 0.7461179
```




```
##
## length of BLUE from geodesic mean : 3.041512 (174 degree)
## length of RED from geodesic mean : -3.100279 (-178 degree)
```



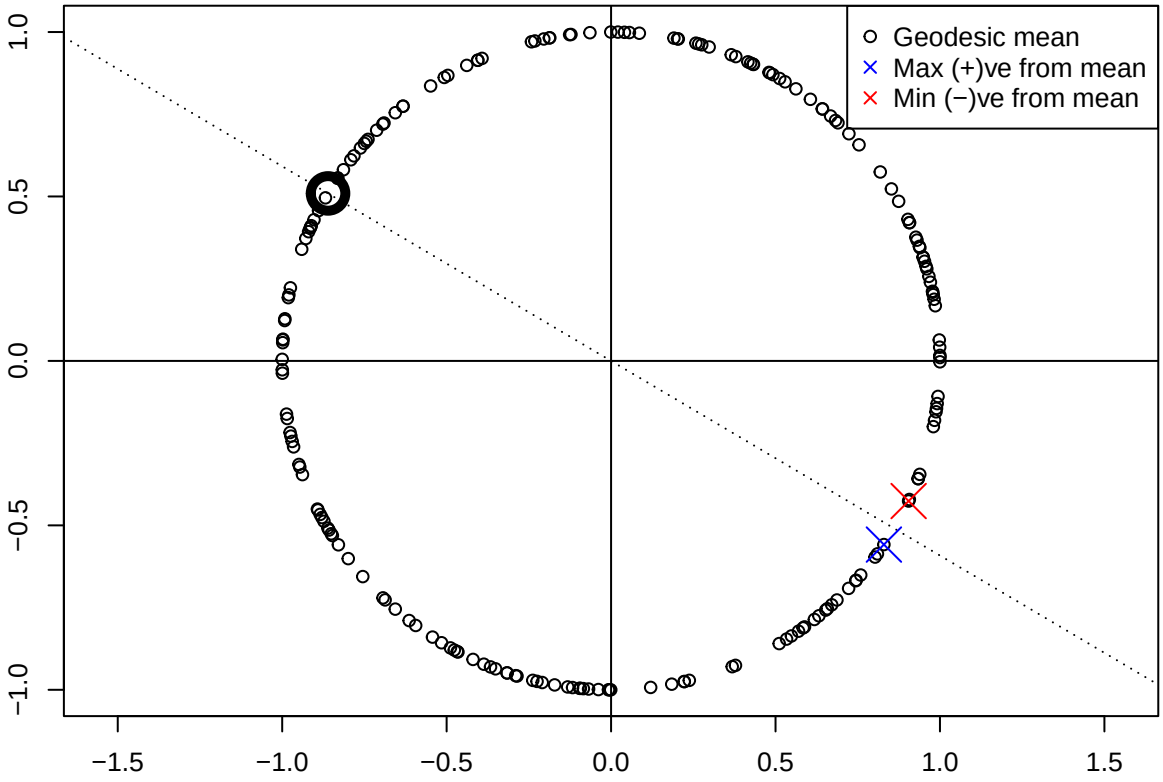

```
## $stress_init
## [1] 0.0220299
##
## $stress_final
## [1] 0.006849383
##
## $rstar
## [1] 1.414214
##
## $r.hat
## [1] 1.427469
```

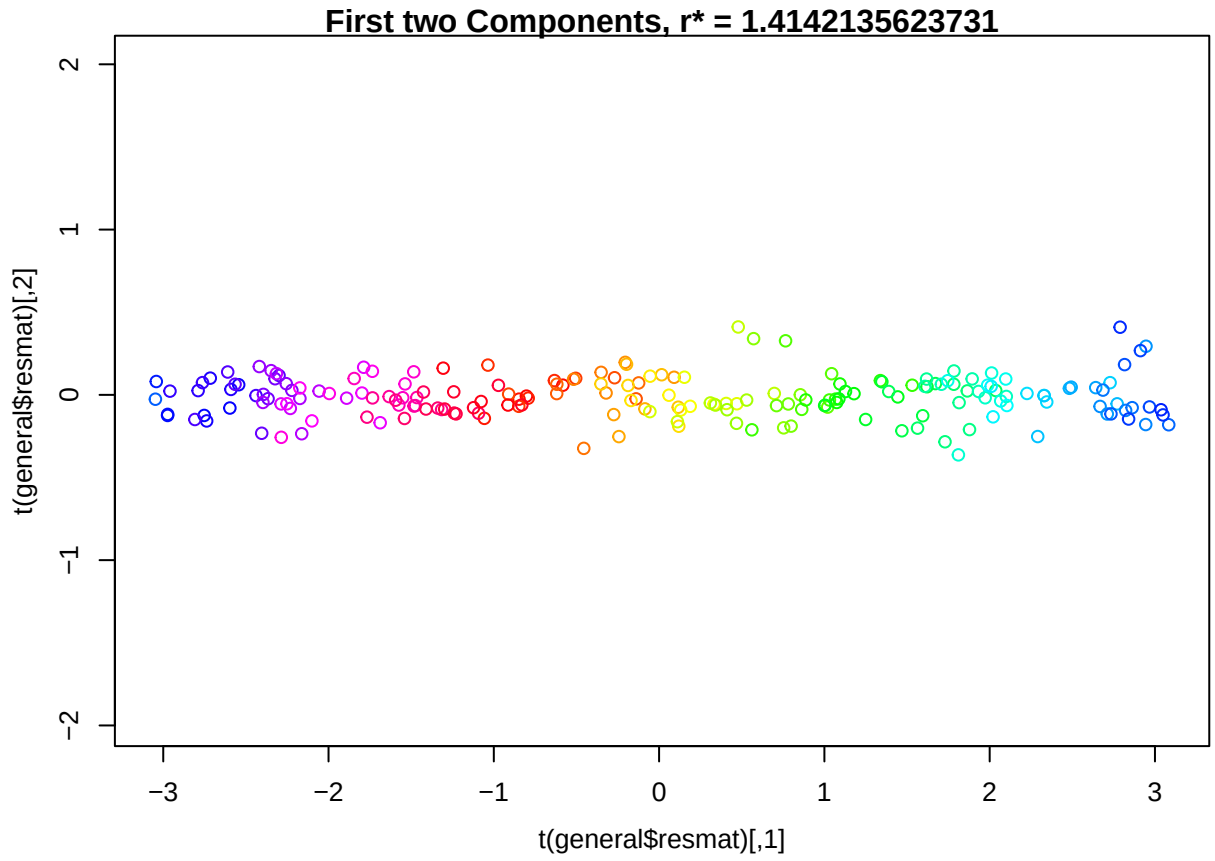



```
print(Sys.time()-time)
```

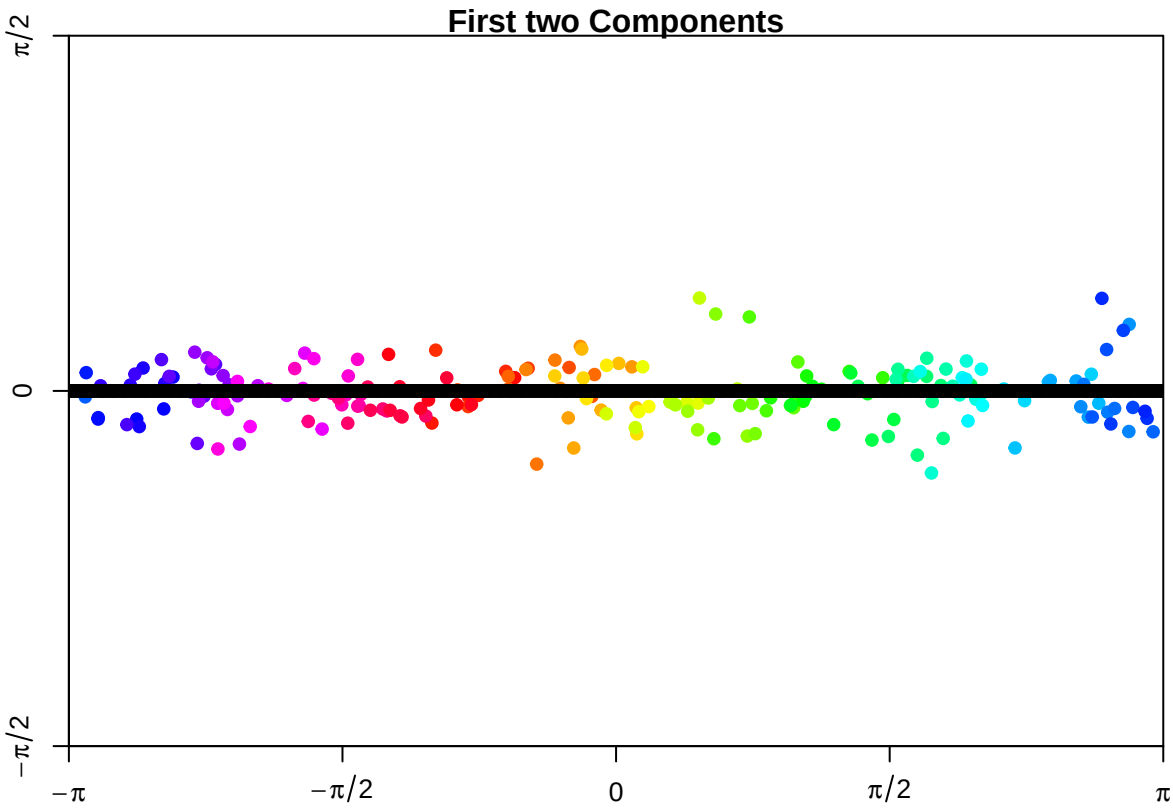
```
## Time difference of 55.63201 secs
```

```
time <- Sys.time()
res <- plot_MDS(clusters, tol = 1e-3)
```


Geodesic mean
Max (+)ve from mean
Min (−)ve from mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5



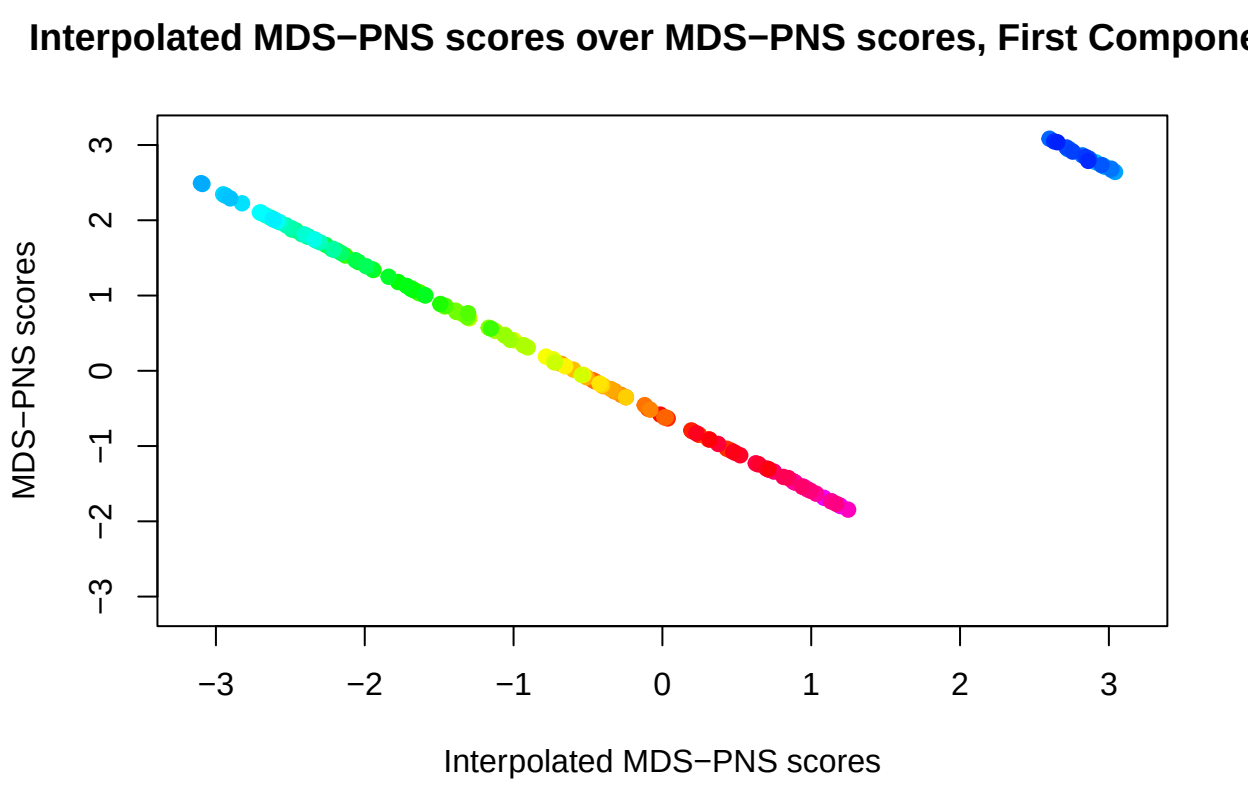

```
## $stress_init
## [1] 0.0220299
##
## $stress_final
## [1] 0.007425575
##
## $rstar
## [1] 1.414214
##
## $r.hat
## [1] 1.425072
```




```
print(Sys.time()-time)
```

```
## Time difference of 53.42579 secs
```

```
res_mod = rep(0, n)
for(i in 1:n){
  diff = res_int[i,1]- res[i,1]
  if (diff < pi){
    res_mod[i] = res[i,1]
  }
  else {
    res_mod[i] = pi - res[i,1]
  }
}
plot(res_int[,1], res_mod,xlab="Interpolated MDS-PNS scores",ylab="MDS-PNS scores",  main = paste("Inte
```



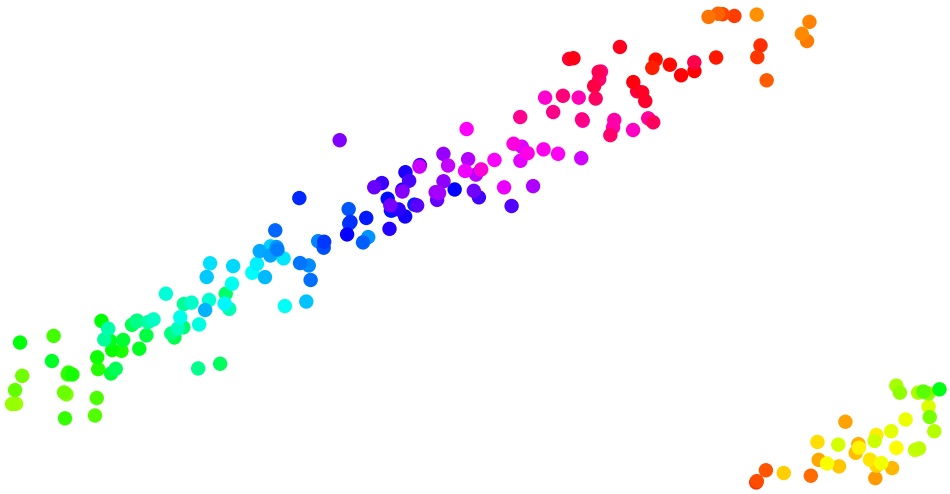

```
ml = lm(res_int[,1] ~ res_mod)
summary(ml)
```

```
##
## Call:
## lm(formula = res_int[, 1] ~ res_mod)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.3425 -1.4513  0.4647  1.1773  2.7600
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.41575    0.14248  -2.918  0.00393 **
## res_mod      0.26394    0.04995   5.284 3.32e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.68 on 198 degrees of freedom
## Multiple R-squared:  0.1236, Adjusted R-squared:  0.1192
## F-statistic: 27.92 on 1 and 198 DF,  p-value: 3.323e-07
```

```
# Straight trends on (S^1)^2
set.seed(7)
n <- 200
```

```
samp_1 <- r_path_s1r(n = n, r = 2, k = c(1, 1), angles = TRUE)

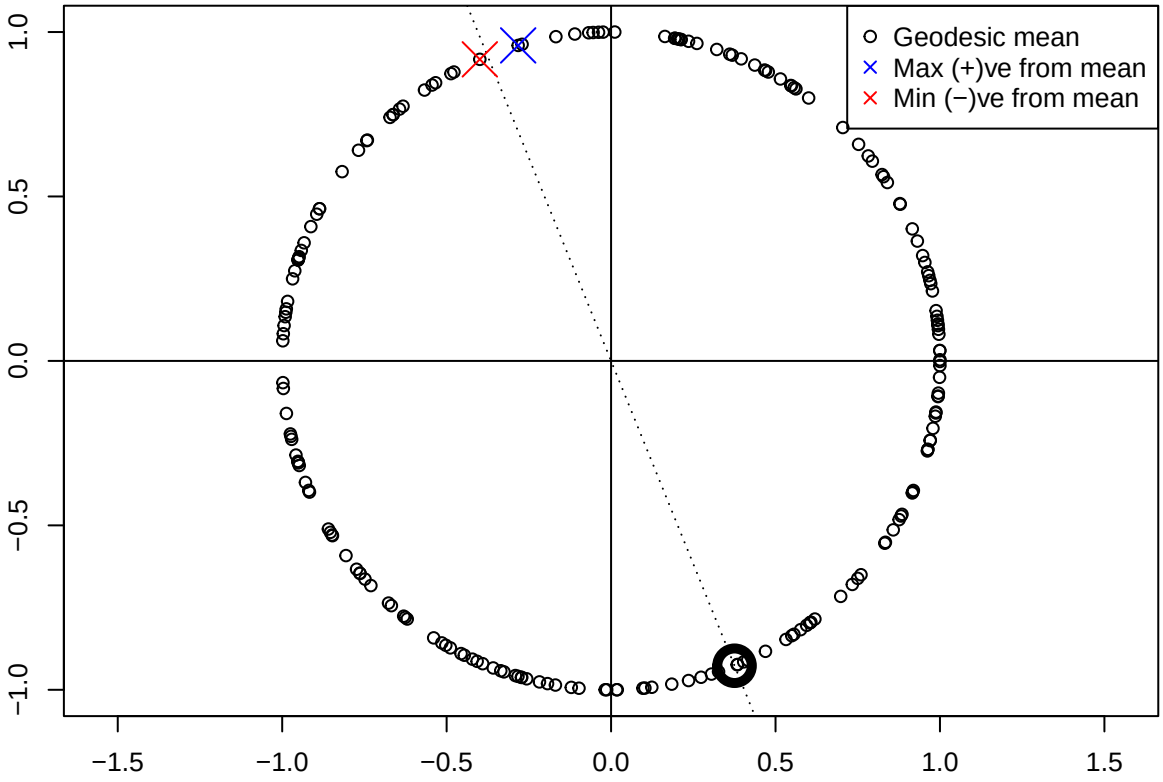
plot(samp_1,
  xlim = c(-pi, pi), ylim = c(-pi, pi), col = rainbow(n),
  axes = FALSE, xlab = "", ylab = "", pch = 16
)
```



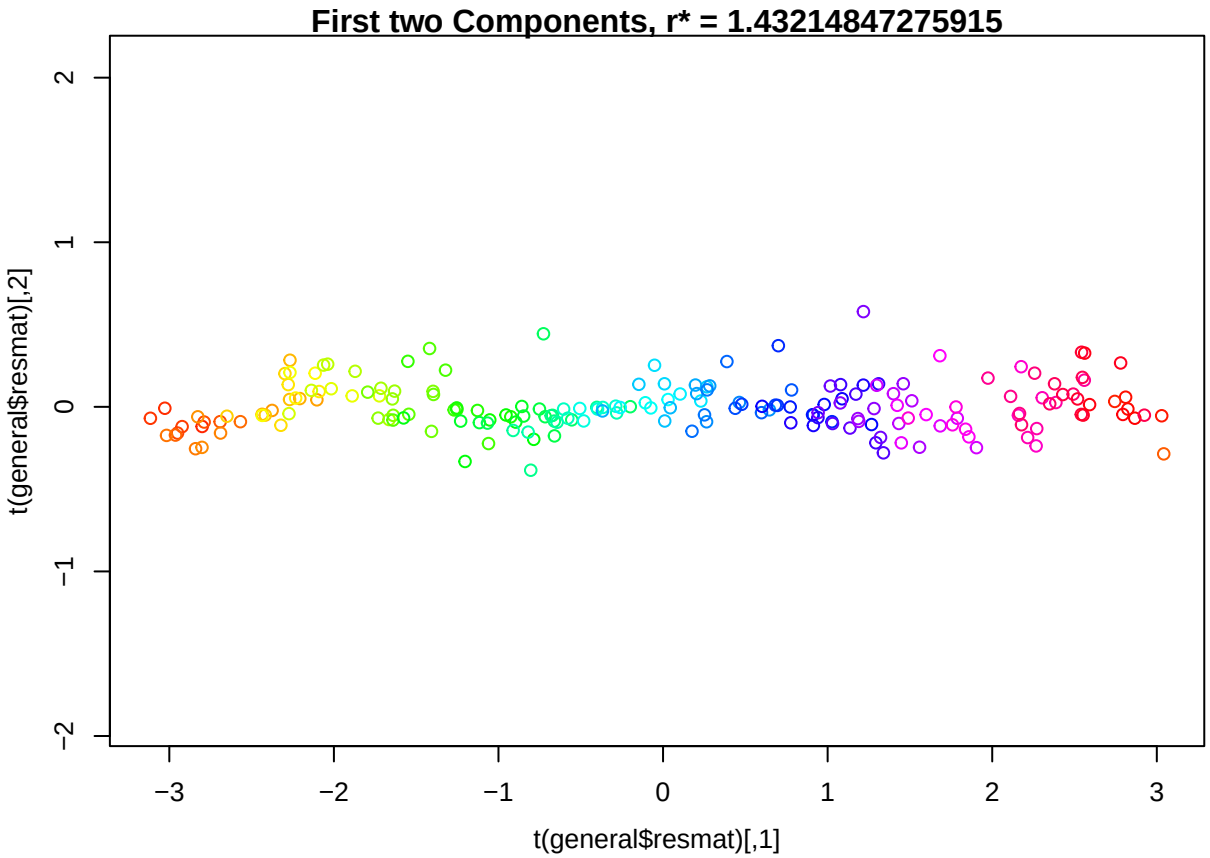
```
clusters <- samp_1

time <- Sys.time()
res_int6 <- plot_MDS_py(clusters, landmark_size = .1,tol=1e-3,landmark_tol=1e-3)
```

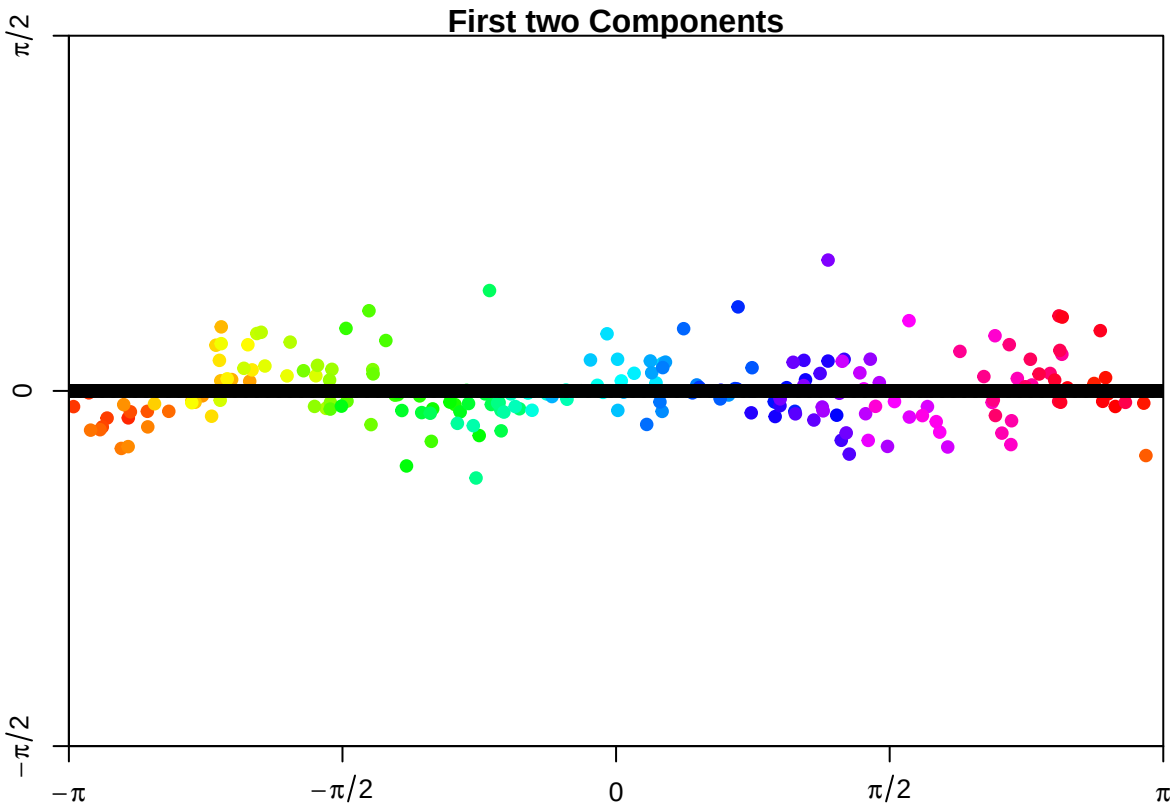
```
## Message from pns() : dataset is on 2-sphere.
##  .. sequential tests with significance level 0.1
## 2-sphere to 1-sphere, by SMALL sphere, p(LRT) = 0.009894088, p(vMF) = 0
```




```
##
## length of BLUE from geodesic mean : 3.043141 (174 degree)
## length of RED from geodesic mean : -3.115899 (-179 degree)
```



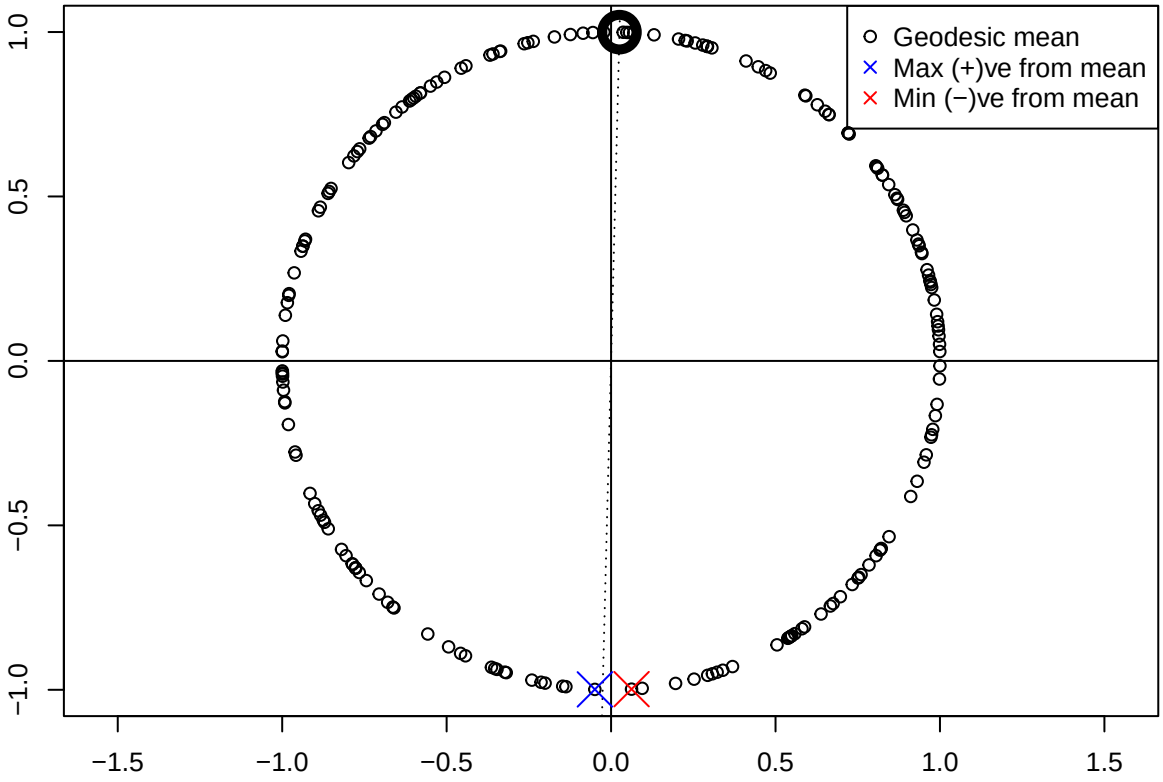

```
## $stress_init
## [1] 0.02483771
##
## $stress_final
## [1] 0.006039672
##
## $rstar
## [1] 1.414214
##
## $r.hat
## [1] 1.432148
```

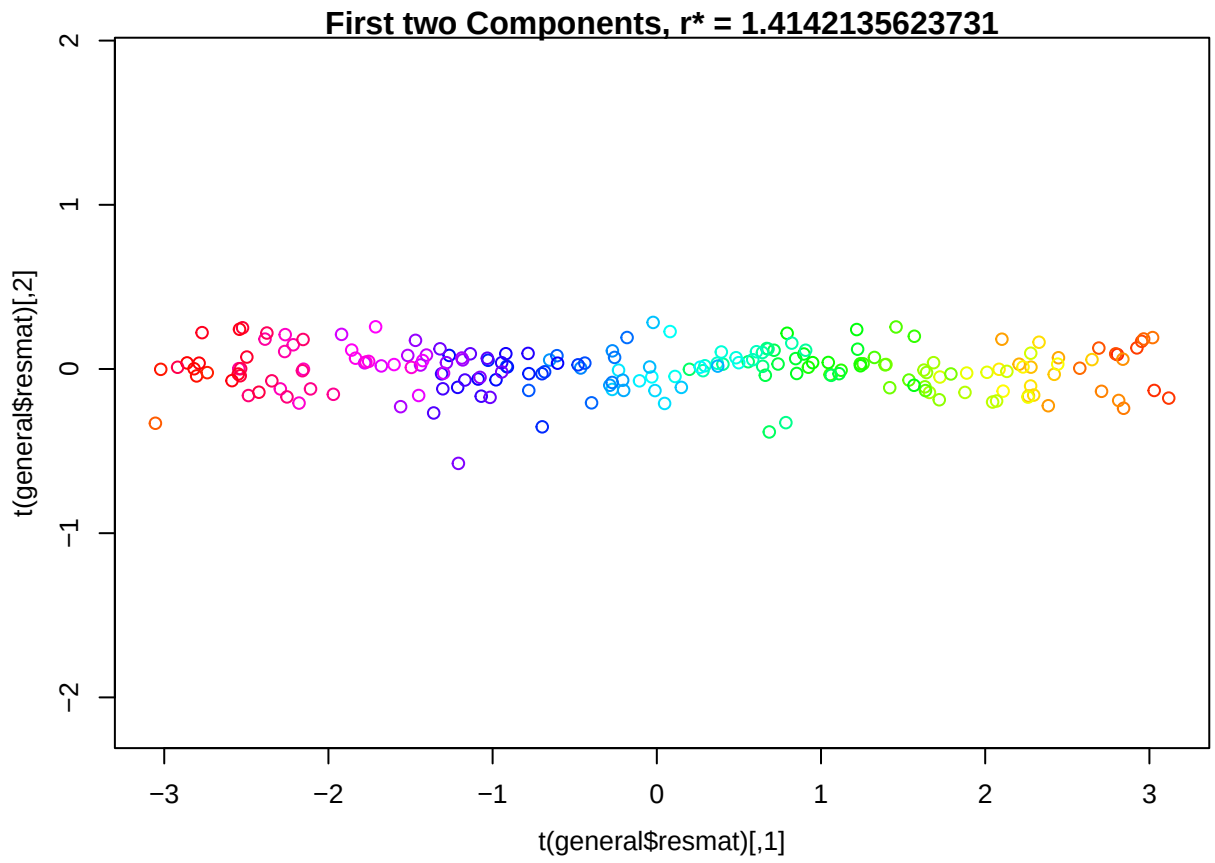



```
print(Sys.time()-time)
```

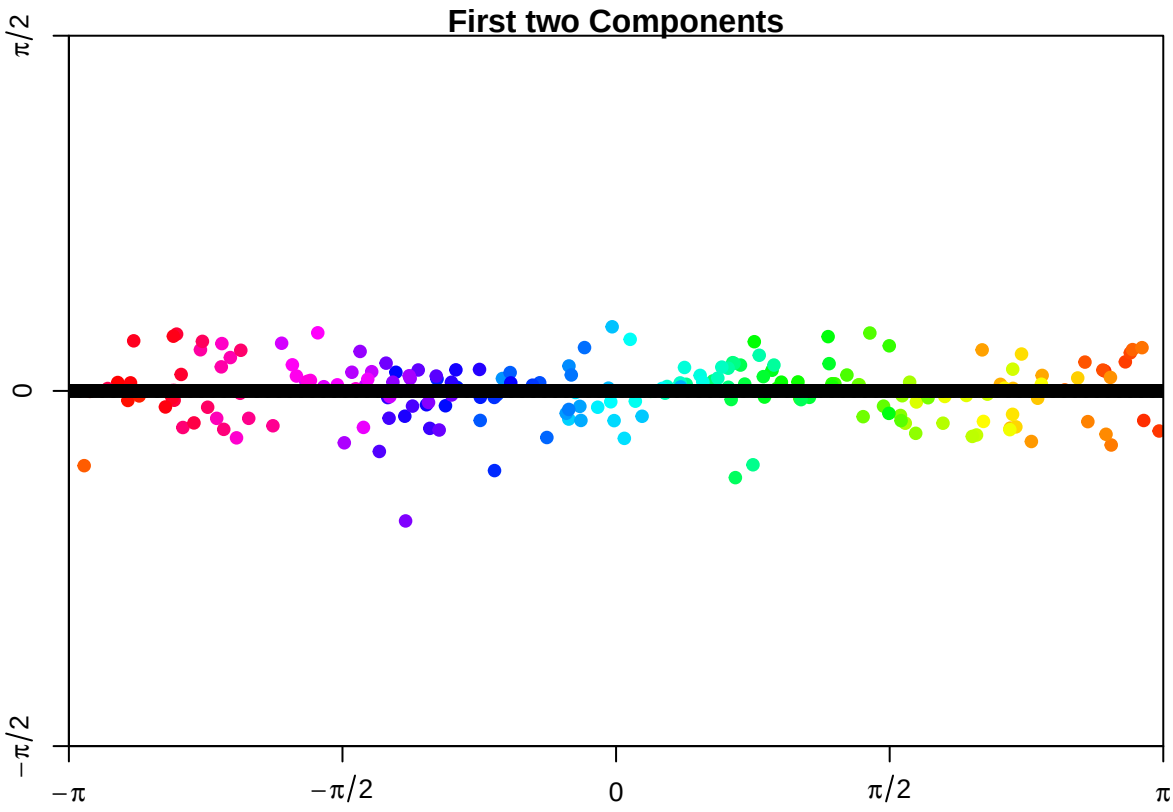
```
## Time difference of 1.853055 mins
```

```
time <- Sys.time()
res6 <- plot_MDS(clusters,tol=1e-3)
```


Geodesic mean
Max (+)ve from mean
Min (−)ve from mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5



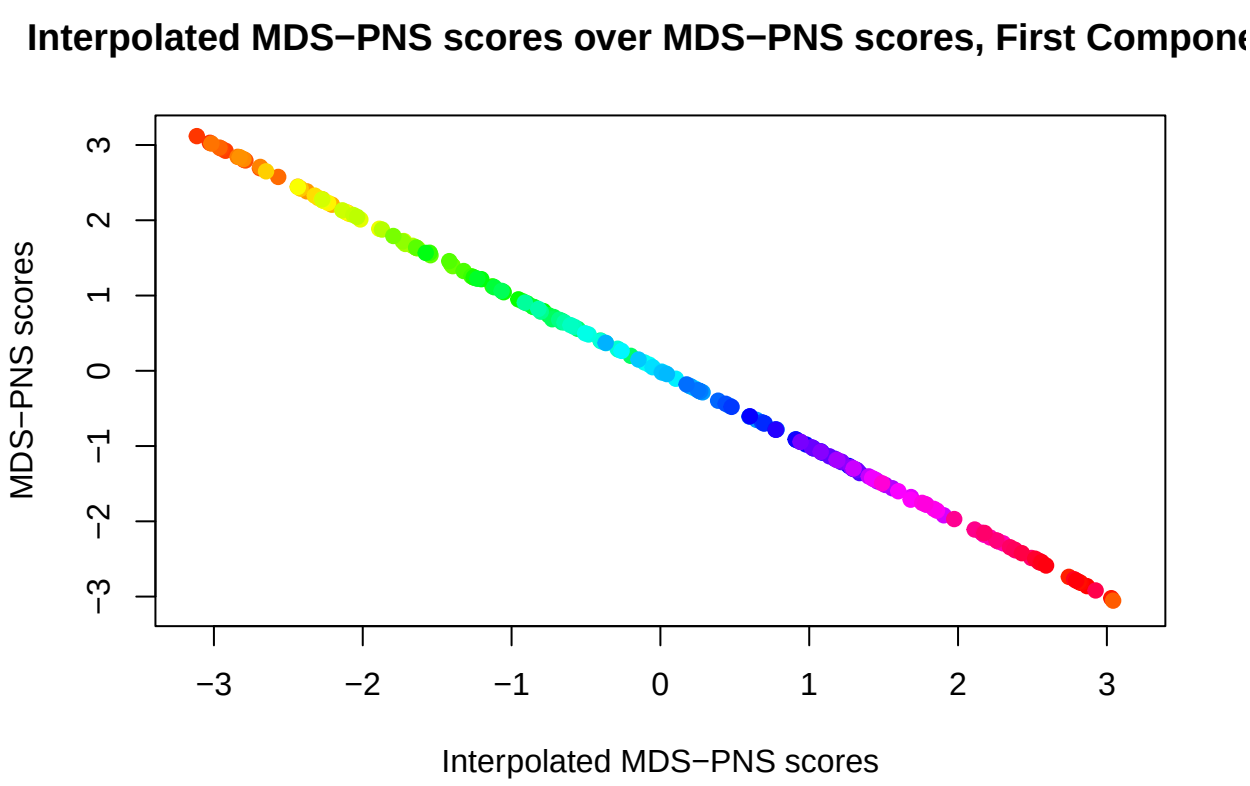

```
## $stress_init
## [1] 0.02483771
##
## $stress_final
## [1] 0.005384824
##
## $rstar
## [1] 1.414214
##
## $r.hat
## [1] 1.426717
```




```
print(Sys.time()-time)
```

```
## Time difference of 59.04658 secs
```

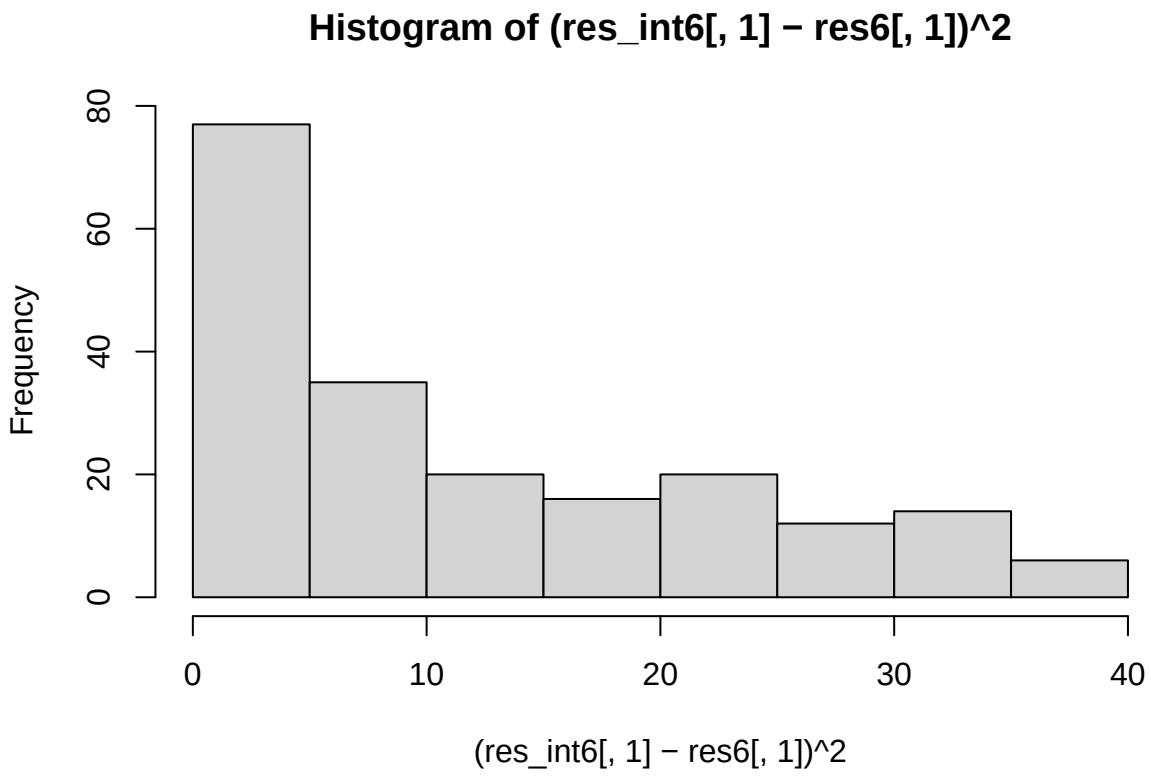
```
plot(res_int6[,1], res6[,1],xlab="Interpolated MDS-PNS scores",ylab="MDS-PNS scores",  main = paste("In
```




```
data <- data.frame(res_int6[,1],res6[,1])
ml = lm(res_int6[,1] ~ res6[,1])
summary(ml)
```

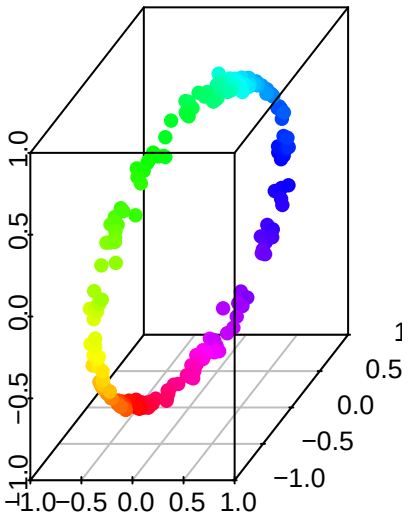
```
##
## Call:
## lm(formula = res_int6[, 1] ~ res6[, 1])
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.041173 -0.004075 -0.000305  0.004088  0.039241
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.790e-15  6.101e-04       0        1
## res6[, 1]   -1.001e+00  3.557e-04   -2813   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.008628 on 198 degrees of freedom
## Multiple R-squared:      1,  Adjusted R-squared:      1
## F-statistic: 7.911e+06 on 1 and 198 DF,  p-value: < 2.2e-16
```

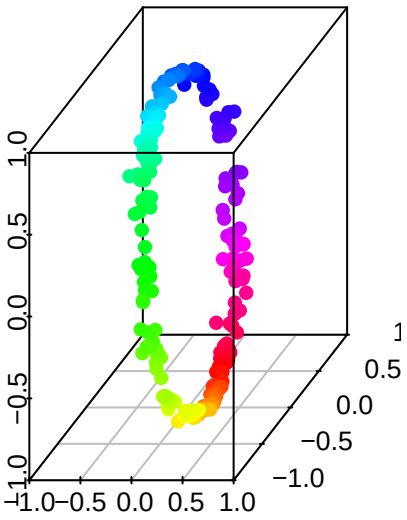
```
hist((res_int6[,1]- res6[,1])^2)
```




```r
#' # Small-circle trends on (S^2)^2
set.seed(20)
n <- 200
samp_3 <- r_path_s2r(n = n, r = 2, sigma = 0.05)
layout(matrix(c(1, 2), 1))

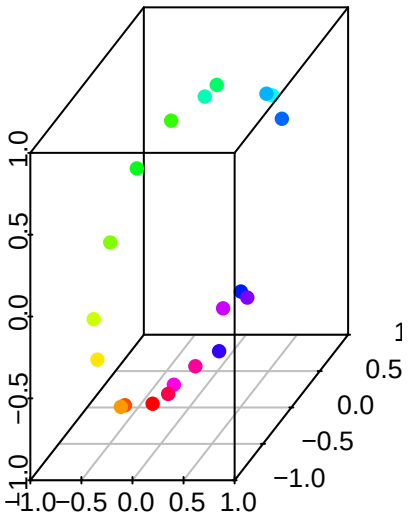
scatterplot3d::scatterplot3d(
  samp_3[, , 1],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(n), pch = 16
)
scatterplot3d::scatterplot3d(
  samp_3[, , 2],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(n), pch = 16
)
```

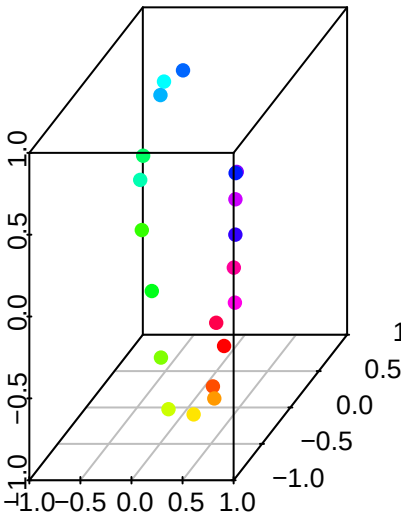




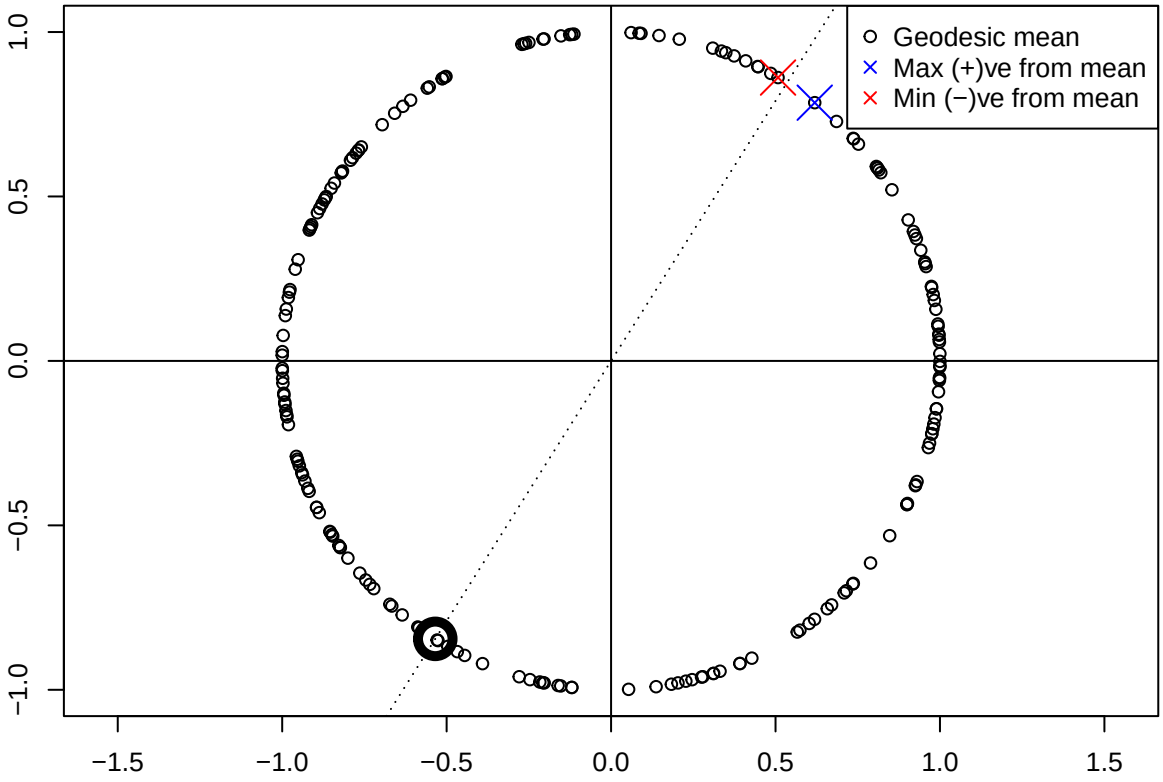

```
centers <- get_center(clusters)

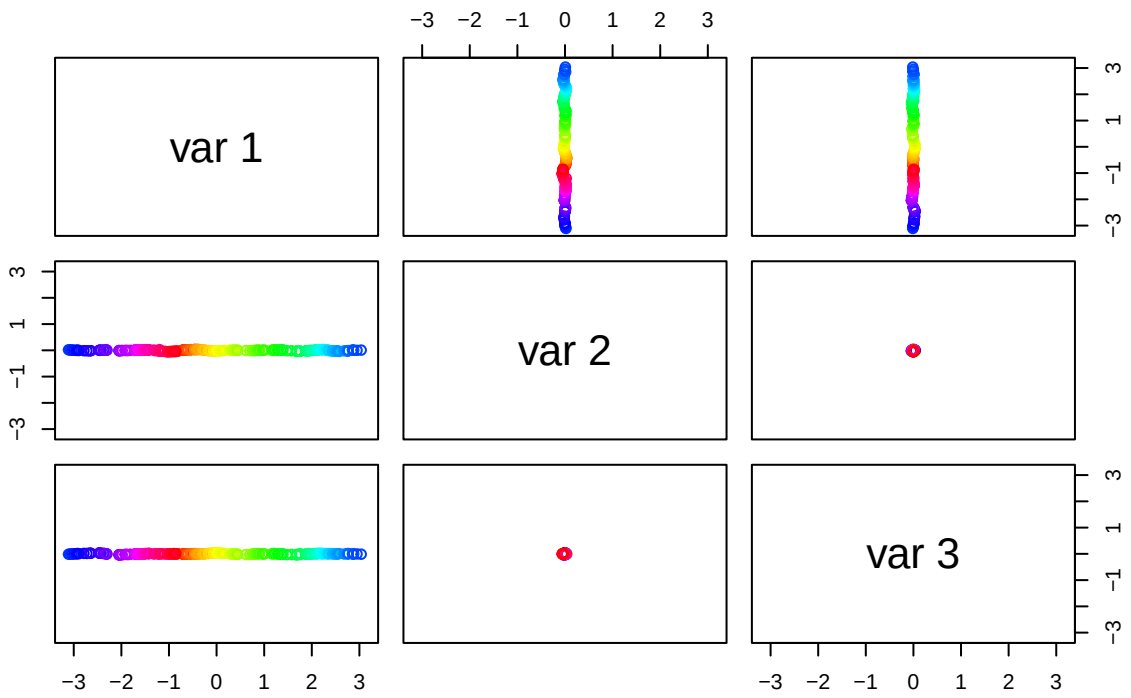
scatterplot3d::scatterplot3d(
  samp_3[centers, , 1],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(length(centers)), pch = 16
)
scatterplot3d::scatterplot3d(
  samp_3[centers, , 2],
  xlim = c(-1, 1), ylim = c(-1, 1), zlim = c(-1, 1),
  xlab = "", ylab = "", zlab = "", color = rainbow(length(centers)), pch = 16
)
```



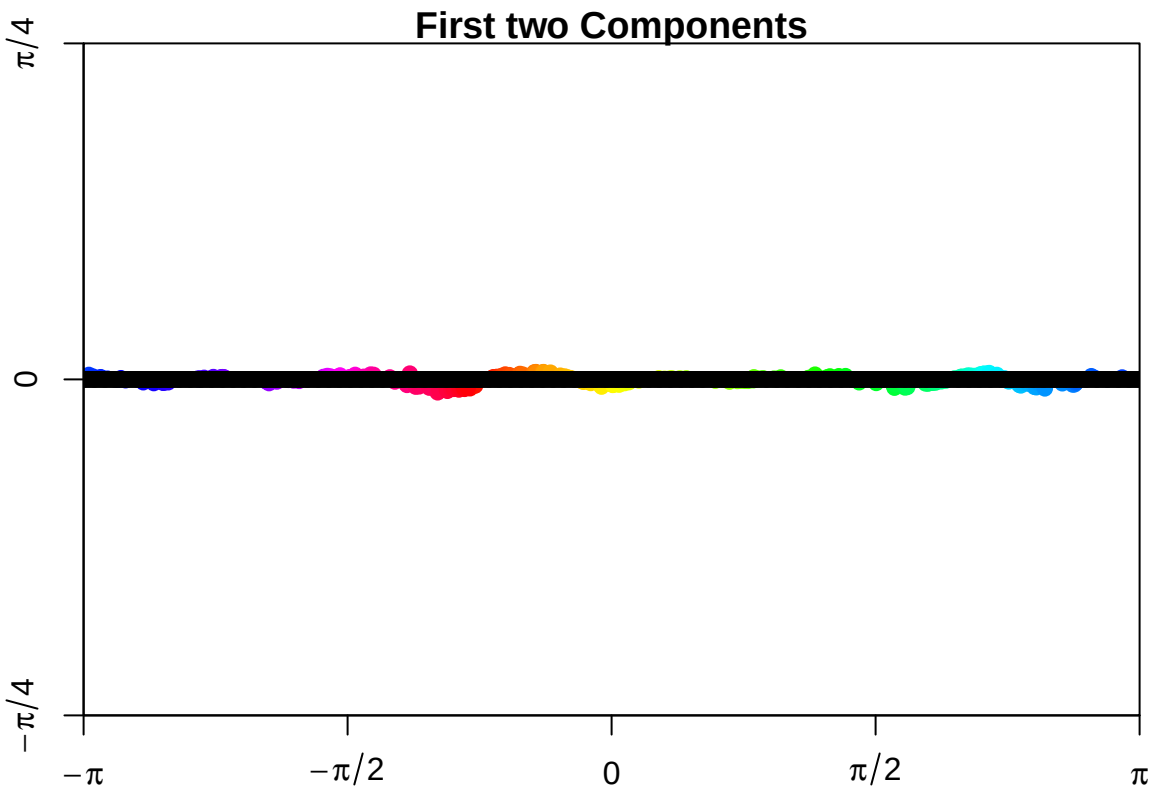



```
clusters <- cbind(samp_3[, , 1], samp_3[, , 2])
time = Sys.time()
pns_data <- plot_MDS(clusters, c(2, 2), angular = FALSE)
```


Geodesic mean
Max (+)ve from mean
Min (−)ve from mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5




```
## $stress_init
## [1] 0.04343599
##
## $stress_final
## [1] 0.0003362942
##
## $rstar
## [1] 1.414214
##
## $r.hat
## [1] 1.411459
```

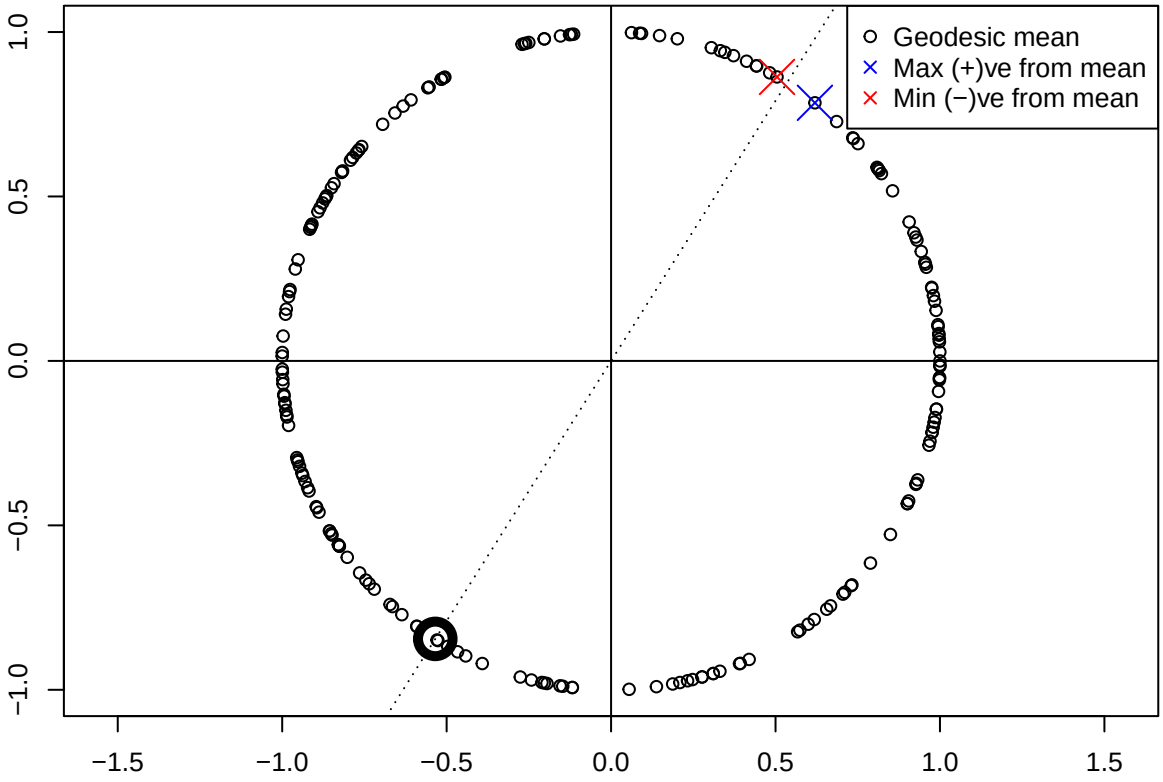



```
print(Sys.time()-time)
```

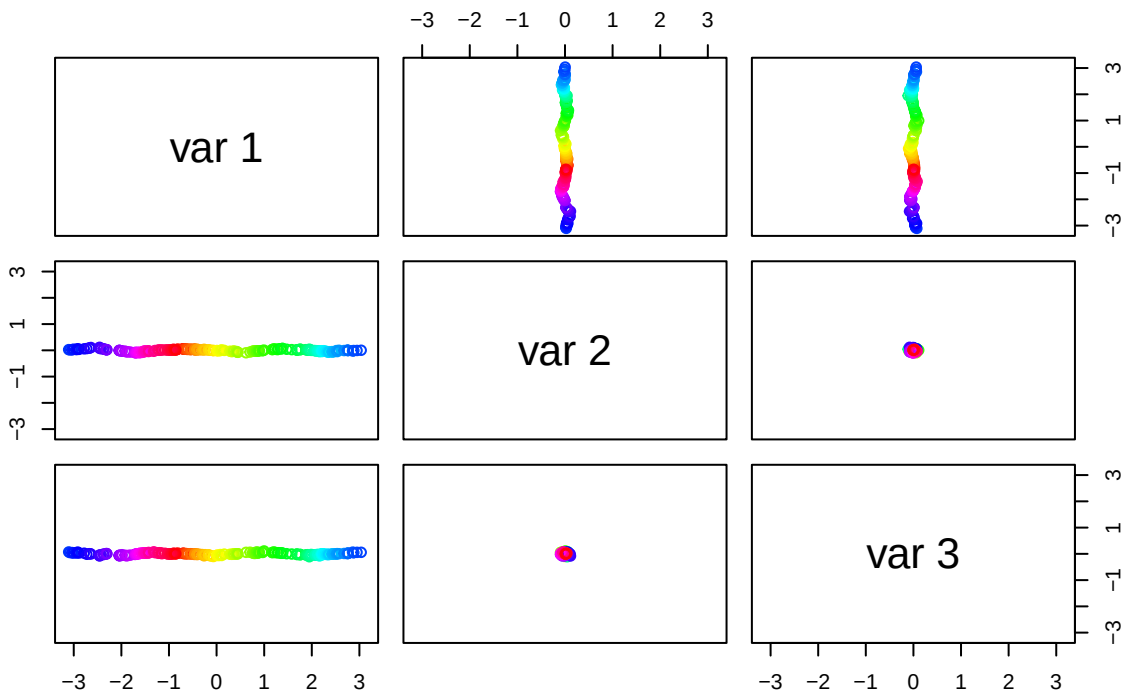
```
## Time difference of 1.519848 mins
```

```
time = Sys.time()
pns_data_int <- plot_MDS_py(clusters, c(2, 2), angular = FALSE, landmark_tol=5e-3)
```

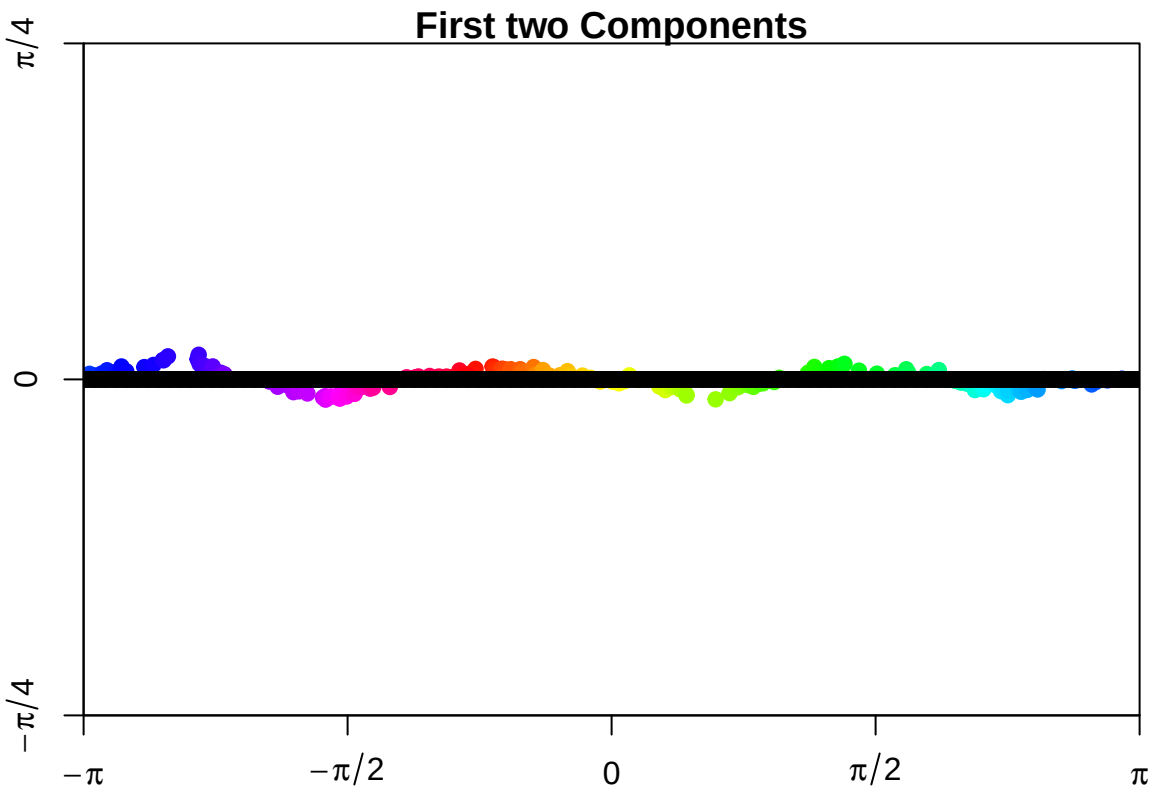
```
## Message from pns() : dataset is on 4-sphere.
##  .. sequential tests with significance level 0.1
## 4-sphere to 3-sphere, by SMALL sphere, p(LRT) = 0.004233561, p(vMF) = 0
## 3-sphere to 2-sphere, by SMALL sphere, p(LRT) = 0.05479871, p(vMF) = 0
## 2-sphere to 1-sphere, by SMALL sphere, p(LRT) = 0.09008832, p(vMF) = 0
```




```
##
## length of BLUE from geodesic mean : 3.036975 (174 degree)
## length of RED from geodesic mean : -3.10657 (-178 degree)
```



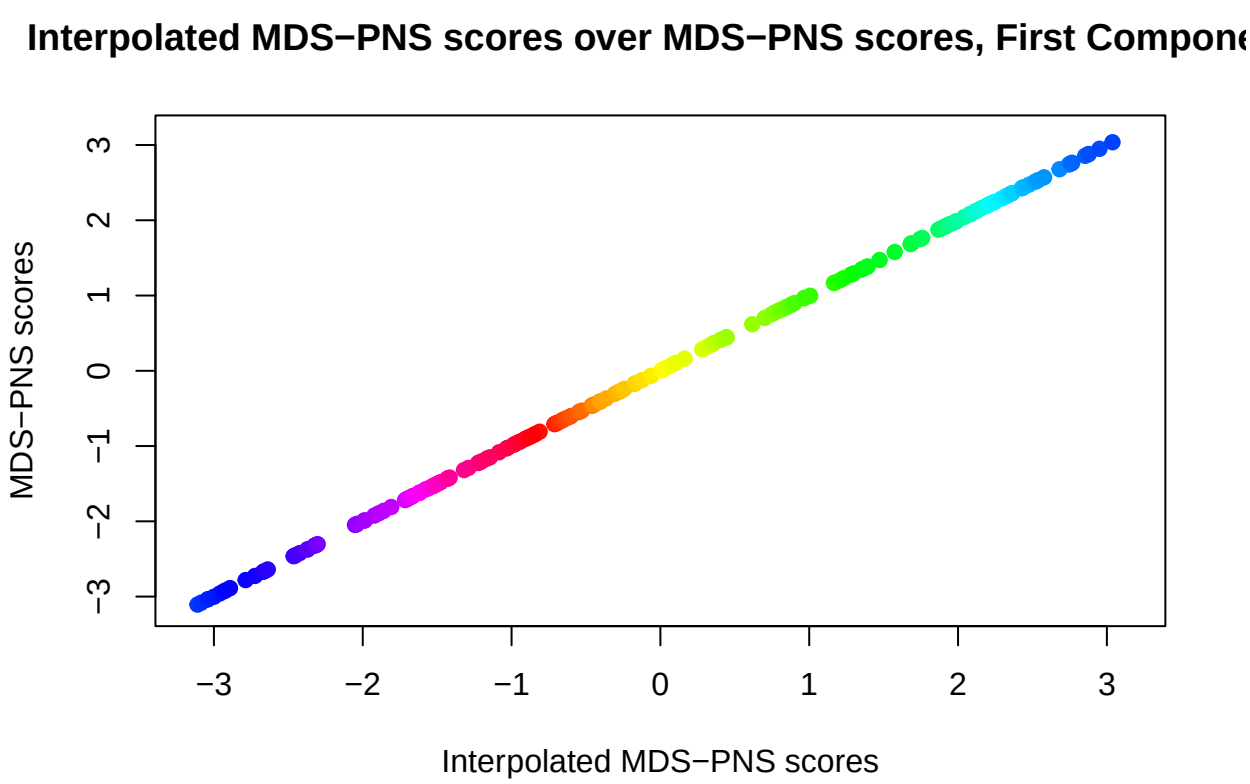

```
## $stress_init
## [1] 0.04343599
##
## $stress_final
## [1] 0.0002839071
##
## $rstar
## [1] 1.414214
##
## $r.hat
## [1] 1.414817
```




```
print(Sys.time()-time)
```

```
## Time difference of 2.480137 mins
```

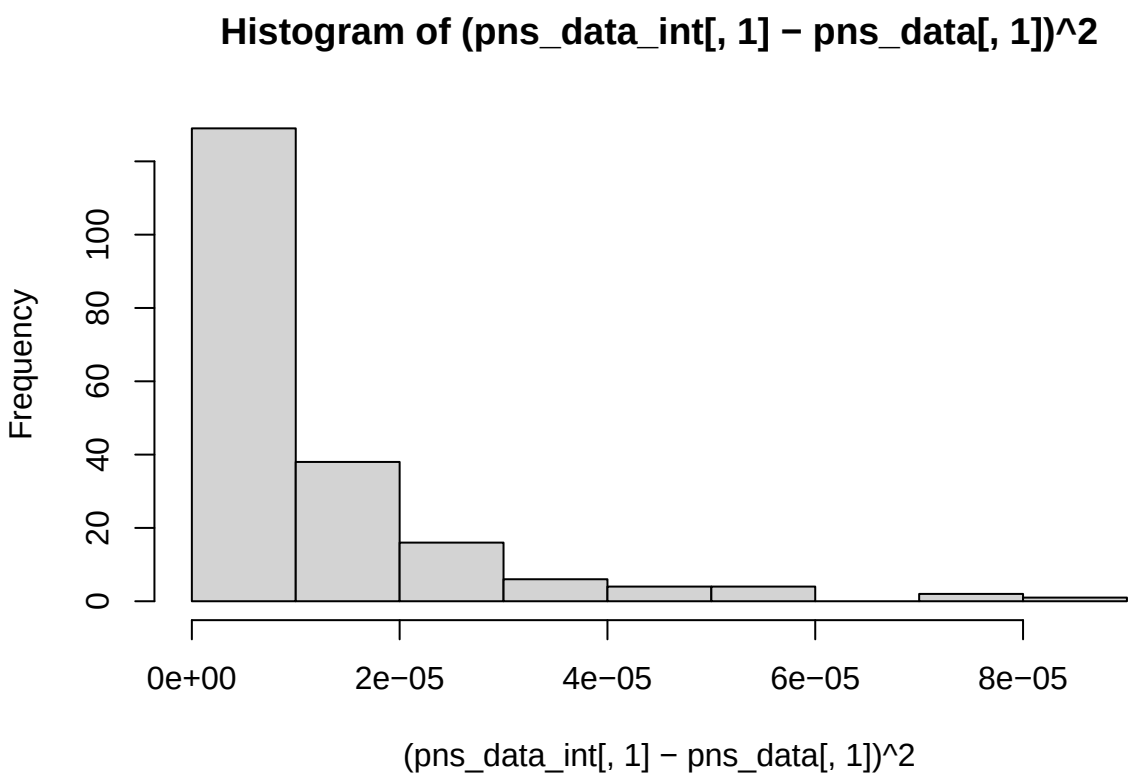
```
plot(pns_data[,1], pns_data_int[,1],xlab="Interpolated MDS-PNS scores",ylab="MDS-PNS scores",  main = pa
```




```
data <- data.frame(pns_data[,1],pns_data_int[,1])
ml = lm(pns_data[,1] ~ pns_data_int[,1])
summary(ml)
```

```
##
## Call:
## lm(formula = pns_data[, 1] ~ pns_data_int[, 1])
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0086673 -0.0021270  0.0002667  0.0021924  0.0090817
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)       3.690e-16  2.301e-04       0        1
## pns_data_int[, 1] 1.000e+00  1.320e-04    7577   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.003254 on 198 degrees of freedom
## Multiple R-squared:      1,  Adjusted R-squared:      1
## F-statistic: 5.74e+07 on 1 and 198 DF,  p-value: < 2.2e-16
```

```
hist((pns_data_int[,1]- pns_data[,1])^2)
```


Histogram of (pns_data_int[, 1] − pns_data[, 1])^2
Frequency
0
20
40
60
80
100
0e+00
2e−05
4e−05
6e−05
8e−05
(pns_data_int[, 1] − pns_data[, 1])^2